

14

NUMBER AND ALGEBRA

POLYNOMIALS

OPTIONAL PATH TOPIC
FOR HSC MATHEMATICS ADVANCED AND EXTENSION

We have analysed and graphed linear equations $y = mx + c$, quadratic equations $y = ax^2 + bx + c$ and simple cubic equations $y = ax^3 + c$. In this topic we will look at equations involving higher powers of x, their graphs and the methods for factorising them and sketching their graphs.



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Chapter outline

	Working mathematically				
14.01 Polynomials*	U	F			C
14.02 Adding and subtracting polynomials*	U	F			
14.03 Multiplying polynomials*	U	F			
14.04 Dividing polynomials*	U	F		R	C
14.05 The remainder theorem*	U	F		R	
14.06 The factor theorem*	U	F		R	C
14.07 Graphing cubic equations $y = a(x - b)(x - c)(x - d)$*	U	F		R	C
14.08 Graphing polynomials*	U	F	PS	R	C
14.09 Transforming graphs of polynomials*	U	F		R	C

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Wordbank

degree of a polynomial The highest power in a polynomial. For example, the degree of $8x^3 + 4x - 7$ is 3

leading coefficient In a polynomial, the number in front of the highest power of the variable, for example, in $8x^3 + 4x - 7$, the leading coefficient is 8

polynomial An algebraic expression involving powers of x that are positive integers. For example, $P(x) = 8x^3 + 4x - 7$

quotient The 'whole' part of the answer when a polynomial is divided by another polynomial

remainder The 'left-over' part of the answer when a polynomial is divided by another polynomial

root of an equation A value of x that is a solution to the equation $P(x) = 0$. A root of $P(x) = 0$ is the same as a *zero* of $P(x)$

zero of a polynomial A value of x that makes the value of the polynomial $P(x)$ equal to 0

Quiz
Wordbank 14

U = Understanding | **F** = Fluency | **PS** = Problem solving | **R** = Reasoning | **C** = Communication

Quiz (1):
- Wordbank 14

Worksheets (2):
14.08 Polynomials review
14.09 Advanced graphs

Puzzles (2):
14.04 Finding the quotient
14.06 Factorising cubic functions

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In this chapter you will:

- (PATH) recognise a polynomial and use polynomial notation
- (PATH) add and subtract polynomials
- (PATH) multiply and divide a polynomial by a linear polynomial
- (PATH) understand and apply the remainder and factor theorems
- (PATH) factorise polynomials and solve polynomial equations
- (PATH) graph cubic equations of the form $y = a(x - b)(x - c)(x - d)$
- (PATH) graph quadratic, cubic and quartic polynomials, including those with multiple factors
- (PATH) graph transformations of a basic polynomial $y = P(x)$

14.01 Polynomials

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A **polynomial** is an algebraic expression involving powers of x that are **positive integers**, for example, $8x^3 + 4x - 7$. It is written using the notation $P(x)$, meaning 'a polynomial using the variable x'.

Polynomials

A **polynomial** has the general form $P(x) = a_n x^n + a_{n-1}x^{n-1} + a_{n-2}x^{n-2} + \ldots + a_2x^2 + a_1x + a_0$, where the powers of x are positive integers, n and a_n, a_{n-1}, a_{n-2}, ..., a_1, a_0 are called **coefficients**.

Terminology	For $P(x) = 2x^5 + 8x^3 - 7$:	Definition
leading term of a polynomial	$2x^5$	The term with the highest power.
leading coefficient of a polynomial	2	The coefficient of the leading term.
degree of a polynomial	5	The highest power of the polynomial.
monic polynomial	Not monic.	A polynomial with a leading coefficient of 1, such as $P(x) = x^4 - 7x^2 + x + 8$
constant term	-7	The term 'at the end' of the polynomial that is independent of x

Names of polynomials

- **Linear polynomials** have degree 1 and their graph is a straight line, for example, $P(x) = 2x - 7$.
- **Quadratic polynomials** have degree 2 and their graph is a parabola, for example, $P(x) = 3x^2 - 2x + 5$.
- **Cubic polynomials** have degree 3 and their graph is a cubic curve, for example $P(x) = 6x^3 - 7x^2 + 9x - 10$.
- **Quartic polynomials** have degree 4, for example $P(x) = -2x^4 + x^3 - 8x^2 - 4x + 13$.

Example 1

State whether each expression is a polynomial.

a $A(x) = \sqrt{2}x^4 - 7x^5 + \frac{3}{x^2}$

b $B(x) = 5x + 2$

c $C(x) = 2\sqrt{x} + x^3 + 7$

d $D(x) = (x^2 - 2)(x + 5)$

Solution

a $A(x) = \sqrt{2}x^4 - 7x^5 + \frac{3}{x^2} = \sqrt{2}x^4 - 7x^5 + 3x^{-2}$ is **not** a polynomial because powers of x must be positive integers and $3x^{-2}$ has a negative power.

b $B(x) = 5x + 2$ is a polynomial.

c $C(x) = 2\sqrt{x} + x^3 + 7 = 2x^{\frac{1}{2}} + x^3 + 7$ is **not** a polynomial because $2x^{\frac{1}{2}}$ has a fractional power.

d $D(x) = (x^2 - 2)(x + 5) = x^3 + 5x^2 - 2x - 10$ is a polynomial.

Example 2

For the polynomial $P(x) = 7x^6 + 5x^3 + 9x - 2$, state:

a the degree

b the leading term

c the leading coefficient

d the constant term

e whether the polynomial is monic.

Solution

a The degree is 6.

b The leading term is $7x^6$.

c The leading coefficient is 7.

d The constant term is –2.

e Since the leading coefficient is not 1, the polynomial is not monic.

$P(x)$ notation

$P(c)$ is the value of a polynomial $P(x)$ when $x = c$. This means that the number c is substituted for x in the polynomial. This is the same as with function notation $f(x)$ from Chapter 13, *Functions*.

PATH

Example 3

For the polynomial $P(x) = 3x^4 - 6x^2 + 4$, find:

a $P(3)$ **b** $P(-2)$ **c** $P(0) - P(-1)$

Solution

a $P(3) = 3(3)^4 - 6(3)^2 + 4$
$= 193$

b $P(-2) = 3(-2)^4 - 6(-2)^2 + 4$
$= 28$

c $P(0) = 3(0)^4 - 6(0)^4 + 4 = 4$
$P(-1) = 3(-1)^4 - 6(-1)^2 + 4 = 1$
$\therefore P(0) - P(-1) = 4 - 1 = 3$

EXERCISE 14.01 ANSWERS ON P. 684

Polynomials U F C

EXAMPLE 1

1 Determine whether each expression is a polynomial. If it is a polynomial, state whether it is monic. C

a $9x^2 - 5x$ **b** $\frac{5}{x^2} + \frac{2}{x^3} + x$ **c** $3x^5 + x^6 - \sqrt{7}$

d $10 - \frac{x}{2}$ **e** $4x + 2\sqrt{x}$ **f** $x^3 + x^2 - 11$

g $\sqrt{5}x^2 - \sqrt{10}x^4 + x^3$ **h** $3\sqrt{x} - 2x^4 + x^5$ **i** $(20 - x)(20x - x^2)$

j $3^x - \frac{x}{3} + x^3$ **k** $(x^2 - 2)^2(x + 3)$ **l** $\frac{9x^3 + x^2 + x}{x}$

EXAMPLE 2

2 For each polynomial, state: C

i the degree **ii** the leading coefficient **iii** the constant term.

a $9x^5 - 2x^4 + 5x^3 + 1$ **b** $7x^4 - 6x^5 + 3$ **c** $11x^2 - 10$

d $-6x$ **e** $8x - \frac{1}{2}x^2 + 7x^5 + 3$ **f** 9

g $4x^3 + \sqrt{2}x^4 - 11 + x^6$ **h** $22 - \frac{5x}{4}$ **i** $\frac{x^3}{3} + \frac{x^2}{2} + x$

EXAMPLE 3

3 If $P(x) = 3x^3 - 2x^2 + x - 1$, evaluate: C

a $P(0)$ **b** $P(2)$ **c** $P(-1)$ **d** $P\left(\frac{1}{2}\right)$ **e** $P\left(\sqrt{2}\right)$

4 If $P(x) = x^2 - x$ and $Q(x) = 1 - x^3$, find: C

a $P(0) + Q(0)$ **b** $P(2) + Q(1)$ **c** $P(4) + Q(-1)$ **d** $P(-2) - Q(-2)$

e $P\left(\frac{1}{2}\right) + Q\left(\frac{1}{4}\right)$ **f** $P\left(2\sqrt{3}\right) + Q\left(\sqrt{3}\right)$ **g** $P[Q(2)]$ **h** $Q[P(2)]$

Adding and subtracting polynomials 14.02

PATH

14.02

Example 4

If $P(x) = x^3 + 3x^2 - 2x - 5$ and $Q(x) = 3x^2 - 5x + 7$, simplify each expression.

a $P(x) + Q(x)$ **b** $P(x) - Q(x)$

Solution

a $P(x) + Q(x) = x^3 + 3x^2 - 2x - 5 + (3x^2 - 5x + 7)$
$= x^3 + 6x^2 - 7x + 2$

b $P(x) - Q(x) = x^3 + 3x^2 - 2x - 5 - (3x^2 - 5x + 7)$
$= x^3 + 3x - 12$

EXERCISE 14.02 ANSWERS ON P. 684

Adding and subtracting polynomials U F

EXAMPLE 4

1 Simplify each expression.

a $(9x^3 + x^2 + x) + (7x^2 + 5x - 2)$ **b** $(3x + 2) - (x^3 - x)$

c $(8x - 2) - (11x + 4x^2)$ **d** $2(x^4 - x^2 + 2) - 3(x^3 + x^2 - x^4)$

e $x^4 - (x^3 - x^2 + 5x)$ **f** $2(5x^4 + x^3 + 2x) + 2(11 - 2x^2 + x^5)$

g $(7x^6 + x^5) + (x^2 + x^3) - (2 - 3x)$

h $8x^4 - x^3 + 6x^2 + 5x - (x + 6) + (x^3 + 9x^2 - 1)$

i $x^4 - 8x^3 + 2x^2 + 7x - 2 - (6x - 3x^2 + 5x^4)$ **j** $6x^3 - 2x^2 - x - (2x - 3) - (3x + 2)$

2 If $P(x) = 4x - 3$ and $Q(x) = x^2 + 7x + 2$, simplify each expression.

a $P(x) + Q(x)$ **b** $P(x) - Q(x)$ **c** $Q(x) - P(x)$ **d** $3P(x) + 2Q(x)$

3 If $A(x) = x^2 - 6$, $B(x) = 11x - 2x^2$ and $C(x) = 9 - 4x$, simplify each expression.

a $C(x) - B(x)$ **b** $A(x) - C(x)$ **c** $B(x) - A(x)$

d $A(x) + B(x) + C(x)$ **e** $A(x) - B(x) + C(x)$ **f** $A(x) - [B(x) + C(x)]$

4 If $R(x) = x^2 - 6x + 5$, $P(x) = x - 1$ and $Q(x) = R(x) - P(x)$, simplify each expression.

a $Q(x)$ **b** $Q\left(3\sqrt{2}\right)$ **c** the values of x if $Q(x) = 0$

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14.03 Multiplying polynomials

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Example 5

If $P(x) = 3x + 1$ and $Q(x) = x^2 - 5x - 2$, find $P(x) \cdot Q(x)$.

Solution

$$\begin{aligned} P(x) \cdot Q(x) &= (3x + 1)(x^2 - 5x - 2) \\ &= 3x(x^2 - 5x - 2) + 1(x^2 - 5x - 2) \\ &= 3x^3 - 15x^2 - 6x + x^2 - 5x - 2 \\ &= 3x^3 - 14x^2 - 11x - 2 \end{aligned}$$

EXERCISE 14.03 ANSWERS ON P. 684

Multiplying polynomials U F

EXAMPLE 5

1 Expand each product.

a $(3x + 2)(x^3 - x)$ **b** $(11x + 4x^2)(8x - 2)$

c $(9x^3 + x^2 + x)(5x - 2)$ **d** $(6x^3 - 2x^2 - x - 3)(3x + 2)$

e $(2 - 3x)(7x^6 + x^5 + x^3 + x^2)$ **f** $(x + 6)(8x^4 - x^3 + 6x^2 + 5x - 2)$

2 If $P(x) = 4x - 3$ and $Q(x) = x^2 + 7x + 2$, find $P(x) \cdot Q(x)$.

3 If $A(x) = x^2 - 6$, $B(x) = 11x - 2x^2$ and $C(x) = 9 - 4x$, find:

a $A(x) \cdot B(x)$ **b** $A(x) \cdot C(x)$ **c** $B(x) \cdot C(x)$

14.04 Dividing polynomials

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Puzzle
Finding the quotient

The **long division** process for $9947 \div 65$ is shown on the right.

- 65 into 99 is 1, remainder 34
- Bring down the 4 from 9947
- 65 into 344 goes 5, remainder 19
- Bring down the 7 from 9947
- 65 into 197 goes 3, remainder 2

$$\begin{array}{r} 153 \\ 65\overline{)9947} \\ -65\downarrow \\ \underline{344} \\ -\ 325\downarrow \\ \underline{197} \\ \underline{195} \\ 2 \end{array}$$

$\therefore 9947 \div 65 = 153$, remainder 2

$$= 153\frac{2}{65}$$

$\therefore 9947 = 65 \times 153 + 2$

9947 is the **dividend**, 65 is the **divisor**, 153 is the **quotient**, 2 is the **remainder**.

The long division process can also be used to divide polynomials.

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For example, $(x^3 + 5x^2 - 6x - 4) \div (x + 3)$ is

- $(x + 3)$ into $(x^3 + 5x^2)$ is x^2, remainder $2x^2$ — $x^3 \div x = x^2$
- Bring down the $(-6x)$ from the dividend
- $(x + 3)$ into $(2x^2 - 6x)$ goes $2x$, remainder $-12x$ — $2x^2 \div x = 2x$
- Bring down the (-4) from the dividend
- $(x + 3)$ into $(-12x - 4)$ goes (-12), remainder 32 — $-12x \div x = -12$

$$
\begin{array}{r}
x^2 + 2x - 12 \\
x+3 \overline{)\, x^3 + 5x^2 - 6x - 4} \\
\underline{x^3 + 3x^2} \\
2x^2 - 6x \\
\underline{2x^2 + 6x} \\
-12x - 4 \\
\underline{-12x - 36} \\
32
\end{array}
$$

$\therefore (x^3 + 5x^2 - 6x - 4) \div (x + 3) = x^2 + 2x - 12$, remainder 32

$\therefore (x^3 + 5x^2 - 6x - 4) = (x + 3)(x^2 + 2x - 12) + 32$

$(x^3 + 5x^2 - 6x - 4)$ is the **dividend**, $(x + 3)$ is the **divisor**, $(x^2 + 2x - 12)$ is the **quotient**, 32 is the **remainder**.

Note that we can express the dividend $P(x)$ as the product of its factors plus the remainder:

$P(x) =$ divisor $\times$ quotient $+$ remainder.

Example 6

Divide $P(x) = x^3 - x + 4$ by $A(x) = x - 3$. Then write $P(x)$ in the form $P(x) = A(x) \cdot Q(x) + R(x)$, where $Q(x)$ is the quotient and $R(x)$ is the remainder.

Solution

$$
\begin{array}{r}
x^2 + 3x + 8 \\
x-3 \overline{)\, x^3 + 0x^2 - x + 4} \\
\underline{x^3 - 3x^2} \\
3x^2 - x \\
\underline{3x^2 - 9x} \\
8x + 4 \\
\underline{8x - 24} \\
28
\end{array}
$$

Write $0x^2$ as there is no x^2 term

$\therefore x^3 - x + 4 = (x - 3)(x^2 + 3x + 8) + 28$

Example 7

Show that $(2x + 5)$ is a factor of $2x^3 + 5x^2 - 2x - 30$.

Solution

If $2x^3 + 5x^2 - 12x - 30 = (2x + 5) \cdot Q(x)$ with no remainder, then $(2x + 5)$ is a factor.

$$
\begin{array}{r}
x^2 \quad -6 \\
2x+5 \overline{)\, 2x^3 + 5x^2 - 12x - 30} \\
\underline{2x^3 + 5x^2} \\
0x^2 - 12x - 30 \\
\underline{-12x - 30} \\
0
\end{array}
$$

The remainder is 0, which means that $2x^3 + 5x^2 - 12x - 30 = (2x + 5)(x^2 - 6)$

$\therefore (2x + 5)$ is a factor of $2x^3 + 5x^2 - 12x - 30$.

PATH

EXERCISE 14.04 ANSWERS ON P. 684

Dividing polynomials UFRC

EXAMPLE 6

1 Perform the following divisions, then write the first polynomial in the form: dividend = divisor × quotient + remainder. R C

a $(x^2 + 7x + 4) \div (x + 2)$ **b** $(x^2 - 6x + 2) \div (x - 3)$

c $(4x^2 + 3x + 10) \div (x - 1)$ **d** $(8x^2 + 9x + 11) \div (2x + 1)$

e $(x^3 + 6x^2 + 5x - 4) \div (x - 3)$ **f** $(4x^3 + 2x^2 + x) \div (x + 4)$

g $(2x^3 - x^2 + 5x + 3) \div (x + 6)$ **h** $(11 - x^2 + 3x^3) \div (x + 2)$

i $(x^5 - x^4 + 8x^3 + 2x^2 - x - 1) \div (x + 1)$ **j** $(x^4 - x^2 - 10) \div (x + 3)$

2 If $P(x) = 3x^2 - 7x + 5$, $W(x) = x^2 + 5x$ and $T(x) = x - 2$, then find the quotient and remainder in each expression. R C

a $P(x) \div T(x)$ **b** $W(x) \div T(x)$

c $[W(x) \times P(x)] \div T(x)$ **d** $[P(x) + W(x)] \div T(x)$

EXAMPLE 7

3 Show that $(2x - 1)$ is a factor of each polynomial. Express $P(x)$ as a product of 2 factors. R C

a $P(x) = 6x^2 + x - 2$ **b** $P(x) = 2x^3 + x^2 + x - 1$

c $P(x) = 8x^2 + 10x - 7$ **d** $P(x) = 6x^3 + x^2 - 1$

e $P(x) = 2x^4 - 7x^3 - 5x^2 + 8x - 2$ **f** $P(x) = 2x^4 - x^3 - 2x^2 + 7x - 3$

g $P(x) = 6x^3 - 3x^2 + 2x - 1$ **h** $P(x) = 11x + x^5 - 6x^2 - 2x^6 - 4$

14.05 The remainder theorem

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The remainder theorem

If a polynomial $P(x)$ is divided by the linear expression $(x - a)$, then the remainder is $P(a)$.

Proof:

Since $(x - a)$ is a polynomial of degree 1:

$P(x) = (x - a) \times Q(x) + R$, where the remainder R is a constant.

Substituting $x = a$, gives:

$P(a) = (a - a) \times Q(a) + R$

$= 0 + R$

$= R$

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Example 8

Find the remainder when $P(x) = 6x^4 - 3x^3 + 2x + 5$ is divided by each linear expression.

a $(x - 2)$ **b** $(x + 3)$

Solution

a Dividing by $(x - 2)$ will give the remainder $P(2)$.

$P(2) = 6(2)^4 - 3(2)^3 + 2(2) + 5$

$= 81$

b Dividing by $(x + 3)$ will give the remainder $P(-3)$.

$P(-3) = 6(-3)^4 - 3(-3)^3 + 2(-3) + 5$

$= 566$

EXERCISE 14.05 ANSWERS ON P. 685

The remainder theorem U F R

EXAMPLE 8

1 Find the remainder when $P(x) = 2x^3 - 3x^2 + x - 1$ is divided by each linear expression.

a $(x - 2)$ **b** $(x + 4)$ **c** $(x - 1)$ **d** $(x - 5)$

e $(x + 1)$ **f** $(x - 10)$ **g** $(x + 3)$ **h** $(x - 3)$

2 Determine the remainder when the first polynomial is divided by the second polynomial.

a $x^2 + 11x - 6, x - 4$ **b** $x^2 - 2x + 3, x - 1$

c $3x^2 - x + 4, x - 2$ **d** $x^3 + x^2 - x, x + 2$

e $3x^3 + 2x^2 - 11, x + 1$ **f** $2x^2 - x - 16, x - 10$

g $5x^3 - x^2 - 6x, x + 1$ **h** $x^6 - 3x^4 + x^3 - 2, x + 2$

i $2x^4 - 3x^2 + 6x - 2, x + 3$ **j** $5x^5 - 2x^4 - 3x^2 + 7x - 8, x + 5$

3 If $P(x) = x^4 - 4x^3 + e$ has a remainder of 7 when it is divided by $(x + 1)$, find the value of e. R

4 When $Q(x) = ax^3 + bx^2 + 2x + 3$ is divided by $(x + 1)$, the remainder is 3. When $Q(x)$ is divided by $(x - 1)$, the remainder is 15. Determine the values of a and b. R

5 Find the values of m and n if the polynomial $P(x) = 3x^6 - 5x^3 + mx^2 + nx + 10$ is divisible by $x + 1$ and $x - 2$. R

Foundation Standard Complex

14.06 The factor theorem

PATH



Puzzle
Factorising cubic functions

If a polynomial $P(x)$ is divided by $(x - a)$ and the remainder is 0, then $(x - a)$ is a factor of $P(x)$.
Furthermore, by the remainder theorem, $P(a) = 0$.

The factor theorem

If $(x - a)$ is a factor of $P(x)$, then $P(a) = 0$.

Factorising polynomials

Example 9

a Show that $(x - 2)$ is a factor of $P(x) = 2x^3 - 7x^2 - 3x + 18$.

b Hence express $P(x)$ as a product of its factors.

Solution

a $P(2) = 2(2^3) - 7(2^2) - 3(2) + 18$

$= 0$

$\therefore (x - 2)$ is a factor of $P(x)$.

b Use long division to find the other factors.

$$
\begin{array}{r}
2x^2-3x-9 \\
x-2\overline{)\,2x^3-7x^2-3x+18} \\
2x^3-4x^2 \\
\hline
-3x^2-3x \\
-3x^2+6x \\
\hline
-9x+18 \\
-9x+18 \\
\hline
0
\end{array}
$$

$\therefore P(x) = (x - 2)(2x^2 - 3x - 9)$

$= (x - 2)(x - 3)(2x + 3)$ Factorising $2x^2 - 3x - 9$

$\therefore 2x^3 - 7x^2 - 3x + 18 = (x - 2)(x - 3)(2x + 3)$

Zeroes of a polynomial

If $(x - a)$ is a factor of $P(x)$, $x = a$ is called a **zero** of the polynomial $P(x)$.

A **zero** of a polynomial $P(x)$ is a value of x that makes $P(x)$ equal to zero.

It is a solution of $P(x) = 0$.

In Example **9** above, $x = 2$ is a **zero** of the polynomial $P(x) = 2x^3 - 7x^2 - 3x + 18$ because $P(2) = 0$.

Example 10

a Factorise $2x^3 - 3x^2 - 29x - 30$.

b Hence solve the equation $2x^3 - 3x^2 - 29x - 30 = 0$.

Solution

a Let $P(x) = 2x^3 - 3x^2 - 29x - 30 = 2(x - a)(x - b)(x - c)$ where a, b, c are constants.

Use 'guess and check' to find a zero of $P(x)$.

Note that if you expanded $2(x - a)(x - b)(x - c)$, then the last term would be the constant term $2(-a)(-b)(-c) = -2abc$.

That means any zero of $P(x)$ must be a factor of the constant term of $P(x)$, which is (-30), because $-2abc = -30$.

The factors of 30 are 1, −1, 2, −2, 3, −3, 5, −5, 6, −6, 10, −10, 15, −15, 30 and −30.

Try these values to find a zero of $P(x)$ by guessing and checking:

$P(1) = 2 - 3 - 29 - 30$

$= -60$

$\neq 0$ $\quad\therefore (x - 1)$ is not a factor

$P(-1) = -2 - 3 + 29 - 30$

$= -6$

$\neq 0$ $\quad\therefore (x + 1)$ is not a factor

$P(-2) = -16 - 12 + 58 - 30$

$= 0$ $\quad\therefore (x + 2)$ is a factor

Now, by long division:

$$
\begin{array}{r}
2x^2 - 7x - 15 \\
x + 2 \overline{)\,2x^3 - 3x^2 - 29x - 30} \\
\underline{2x^3 + 4x^2} \\
-7x^2 - 29x \\
\underline{-7x^2 - 14x} \\
-15x - 30 \\
\underline{-15x - 30} \\
0
\end{array}
$$

$\therefore P(x) = (x + 2)(2x^2 - 7x - 15)$

$= (x + 2)(x - 5)(2x + 3)$

$\therefore 2x^3 - 3x^2 - 29x - 30 = (x + 2)(x - 5)(2x + 3)$

b $2x^3 - 3x^2 - 29x - 30 = 0$

$(x + 2)(x - 5)(2x + 3) = 0$

$x + 2 = 0$ or $x - 5 = 0$ or $2x + 3 = 0$

$x = -2$ $\qquad x = 5$ $\qquad 2x = -3$

$x = -1\frac{1}{2}$

$\therefore x = -2, x = 5$ or $x = -1\frac{1}{2}$

PATH

EXERCISE 14.06 ANSWERS ON P. 685

The factor theorem U F R C

EXAMPLE 9

1 For each polynomial $P(x)$, test whether each linear polynomial, **A**, **B** and **C**, is a factor.

	$P(x)$	A	B	C
a	$x^2 - 8x + 7$	$(x - 2)$	$(x - 7)$	$(x - 1)$
b	$x^3 + 2x^2 - x - 2$	$(x + 3)$	$(x + 1)$	$(x - 1)$
c	$2x^3 + x + 3$	$(x + 1)$	$(x - 2)$	$(x - 3)$
d	$x^3 - 5x^2 - 22x - 16$	$(x + 1)$	$(x + 2)$	$(x - 8)$
e	$x^3 + 2x^2 - 13x + 10$	$(x + 5)$	$(x - 2)$	$(x - 3)$

2 Show that the second polynomial is a factor of the first polynomial.

a $x^2 + 10x - 24, x - 2$

b $x^3 + 5x^2 + 11x + 10, x + 2$

c $x^3 + 3x^2 - x - 3, x - 1$

d $x^4 - x^3 + 2x^2 - x - 1, x - 1$

e $x^3 - x^2 - 12x, x + 3$

f $x^5 - 4x^4 + 3x^3 + x^2 + 4, x - 2$

g $8x^4 + 2x^2 - 1, x - \frac{1}{2}$

h $2x^4 + 7x^3 - 56x^2 - 37x + 84, x - 4$

EXAMPLE 10

3 Factorise each polynomial. R C

a $x^3 + 6x^2 + 8x$

b $x^3 - x^2 - 2x$

c $x^3 + 2x^2 - x - 2$

d $2x^3 + 3x^2 - 18x + 8$

e $x^3 - 6x^2 + 11x - 6$

f $x^3 + x^2 - 46x + 80$

g $3x^3 - 16x^2 - 13x + 6$

h $6x^3 - 13x^2 + x + 2$

i $2x^4 - 5x^3 + 2x^2$

j $2x^4 - 3x^3 - 18x^2 + 17x + 30$

4 Solve each equation.

a $(2x - 1)(x - 3)(x + 4) = 0$

b $(x + 4)(x^2 + 1) = 0$

c $(2x - 5)(x^2 - x - 6) = 0$

d $(x^2 - 25)(x^2 + 3) = 0$

e $(x - 2)(x + 3)(x^2 - 16) = 0$

f $x^3 + 5x^2 - 14x = 0$

g $x^3 + 2x^2 - 9x - 18 = 0$

h $x^3 - x^2 - 10x - 8 = 0$

i $x^3 - 21x - 20 = 0$

j $2x^3 - 7x^2 + 2x + 3 = 0$

k $12x^3 - 23x^2 + 13x - 2 = 0$

l $x^3 - x - 24 = 0$

5 Find the value of k if: R

a $x^2 + kx - 5$ is divisible by $x + 5$

b $x - 3$ is a factor of $x^3 - kx^2 - 5x + 6$

6 Factorise $p(x) = 2x^5 - 22x^4 + 78x^3 - 90x^2$ fully and hence solve the equation $2x^5 - 22x^4 + 78x^3 - 90x^2 = 0$. R

Foundation Standard Complex

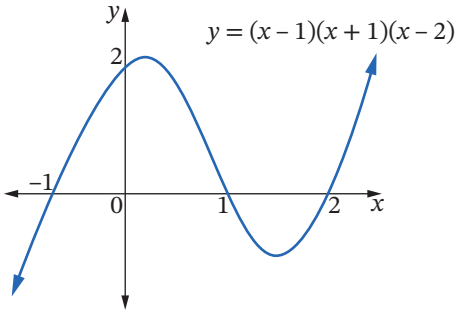
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Graphing cubic equations $y = a(x - b)(x - c)(x - d)$

14.07

PATH

$y = (x - 1)(x + 1)(x - 2)$ is a **cubic equation** because if you expand the RHS, you get $y = x^3 - 2x^2 - x + 2$, and the first term is x^3. Its graph is a more complicated cubic curve:



$y = a(x - b)(x - c)(x - d)$ is called the **intercept form of a cubic equation**. Looking at the graph of $y = (x - 1)(x + 1)(x - 2)$ above, can you see why?

In Chapter 2, *Graphing lines*, we graphed the line $y = mx + c$ by using its x- and y-intercepts and substituting $y = 0$ and $x = 0$ respectively. We can graph the cubic curve $y = a(x - b)(x - c)(x - d)$ in a similar way.

Example 11

Graph the cubic equation $y = x(x + 5)(x - 1)$.

Solution

Substitute $y = 0$ to find the x-intercepts.

$0 = x(x + 5)(x - 1)$

This is true if $x = 0$ or $x + 5 = 0$ or $x - 1 = 0$

If 3 numbers multiplied together equal 0, then one of them must be 0.

$\therefore x = 0, -5$ or 1

The x-intercepts are -5, 0 and 1.

Substitute $x = 0$ to find the y-intercept.

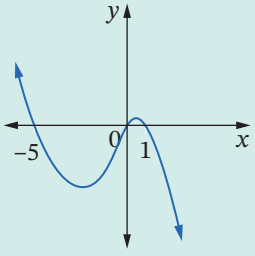
$y = 0(0 + 5)(0 - 1)$

$= 0$

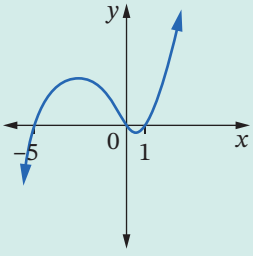
The y-intercept is 0.

Possible graphs are:

A



B



PATH

To determine which graph is correct, we look at a, the coefficient of the cubic equation $y = a(x - b)(x - c)(x - d)$.

In the expansion of $y = x(x + 5)(x - 1)$, the coefficient of x^3 is 1, which is positive, so the correct graph is B, an increasing cubic curve.

(If the coefficient of x^3 is negative, the correct graph is A, a decreasing cubic curve)

We can check this by substituting a value of x, say $x = -3$, into the equation to find a point on the curve:

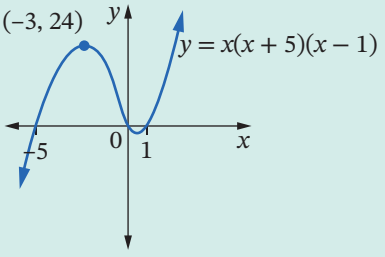
$y = -3(-3 + 5)(-3 - 1)$

$= 24$

This means $(-3, 24)$ lies on the curve.

So B must be the correct graph.

Use graphing technology to check the shape of this graph.



The cubic curve $y = a(x - b)(x - c)(x - d)$

- The cubic curve $y = a(x - b)(x - c)(x - d)$ has x-intercepts of b, c and d.
- If a is positive, the graph is increasing, but decreasing in the middle.
- If a is negative, the graph is decreasing, but increasing in the middle.

EXERCISE 14.07 ANSWERS ON P. 685

Graphing cubic equations $y = a(x - b)(x - c)(x - d)$ U F R C

Graphing technology can be used to check the shapes of the graphs in this exercise.

EXAMPLE 11

1 For each cubic equation, find the x- and y-intercepts and sketch its graph. R

a $y = (x - 2)(x + 2)(x + 3)$ **b** $y = x(x - 2)(x + 1)$

c $y = (x - 1)(x - 3)(x + 1)$ **d** $y = (4 - x)(x + 1)(x - 5)$

e $y = (1 - x)(2 + x)(x + 3)$ **f** $y = 2x(x - 6)(x - 3)$

g $y = -(x + 3)(x - 2)(x - 5)$ **h** $y = -2(x - 1)(x + 2)(x - 3)$

i $y = (x + 2)(x - 1)(x - 2)$

2 What are the x-intercepts of the graph of $y = 2x(x + 1)(x - 3)$?
Select the correct answer **A**, **B**, **C** or **D**.

A $x = 2, 1, -3$ **B** $x = 0, -1, 3$ **C** $x = 0, 1, -3$ **D** $x = 2, -1, 3$

3 Which cubic equation has a graph with x-intercepts -2, -5 and 6? Select **A**, **B**, **C** or **D**.

A $y = (x + 2)(x + 5)(x - 6)$ **B** $y = (x - 2)(x - 5)(x + 6)$

C $y = (x + 2)(x - 5)(x + 6)$ **D** $y = (x - 2)(x + 5)(x + 6)$

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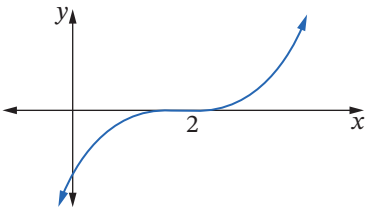
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4 Match each polynomial with its graph. R C

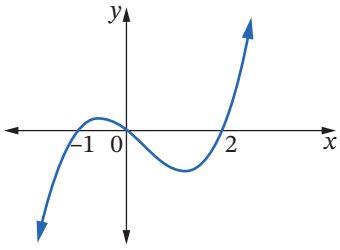
a $y = x(x-2)(x+1)$ **b** $y = (x+1)^2(x+3)$

c $y = -x(x-3)(x+1)$ **d** $y = (x-2)^3$

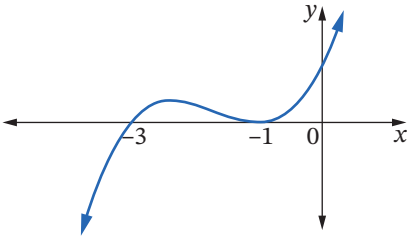
A



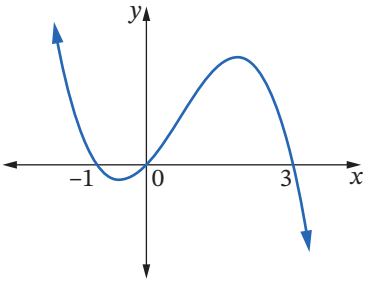
B



C



D



5 **a** Prove that the y-intercept of $y = a(x-b)(x-c)(x-d)$ is $-abcd$. R C

b Prove that the x-intercepts of $y = a(x-b)(x-c)(x-d)$ are b, c and d.

6 Sketch the graph of each cubic equation. R

a $y = x(x-3)(x+1)$ **b** $y = -(x-3)(x-1)(x+1)$

c $y = -2x(x-1)(x+6)$ **d** $y = (2x-3)(x-1)(x+2)$

e $y = (x+1)(x+1)(x+1)$ **f** $y = (x-3)^2(x+2)$

PATH

14.08

Graphing polynomials

14.08

Graphing polynomials

To graph the polynomial $y = P(x)$:

- substitute $y = 0$ to find the x-intercepts
- substitute $x = 0$ to find the y-intercepts
- use the sign of the leading coefficient to sketch the shape of the curve

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Worksheet
Polynomials review

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Example 12

Sketch $y = x^3 - x^2 - 10x - 8$.

Solution

Substitute $y = 0$ to find the x-intercepts.

$0 = x^3 - x^2 - 10x - 8$

Use the factor theorem to factorise the RHS.

Let $P(x) = x^3 - x^2 - 10x - 8$

Test factors of -8.

$P(1) = 1 - 1 - 10 - 8$

$= -18$

$\neq 0$

$P(-1) = (-1)^3 - (-1)^2 - 10(-1) - 8$

$= 0$

$\therefore (x + 1)$ is a factor.

$$
\begin{array}{r}
x^2 - 2x - 8 \\
x+1 \overline{)\, x^3 - x^2 - 10x - 8} \\
\underline{x^3 + x^2} \\
-2x^2 - 10x \\
\underline{-2x^2 - 2x} \\
-8x - 8 \\
\underline{-8x - 8} \\
0
\end{array}
$$

$\therefore P(x) = (x + 1)(x^2 - 2x - 8)$

$= (x + 1)(x - 4)(x + 2)$

If $P(x) = 0$: $(x + 1)(x - 4)(x + 2) = 0$,

$\therefore$ the x-intercepts are -2, -1 and 4.

Substitute $x = 0$ to find the y-intercept.

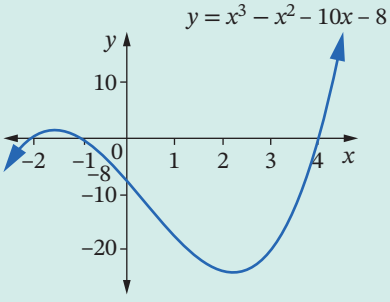
$y = 0^3 - 0^2 - 10(0) - 8$

$= -8$

$\therefore$ The y-intercept is -8.

The leading coefficient is 1, which is positive, so the cubic curve is increasing at the end.

Use graphing technology to check the shape of this graph.



Single, double and triple roots of a polynomial equation

If $x = a$ is a solution of the polynomial equation $P(x) = 0$, then $x = a$ is called a **root** of the equation.

Root is another name for **solution.**

Roots of P(x) = 0	Example
If $(x - a)$ is a factor of a polynomial $P(x)$, then $x = a$ is a **single root** of $P(x) = 0$.	$H(x) = (x - 1)(x + 2)(x - 3)$. $x = 1$, $x = -2$, $x = 3$ are all single roots of $H(x) = 0$.
If $(x - a)^2$ is a factor of a polynomial $P(x)$, then $x = a$ is a **double root** of $P(x) = 0$.	$P(x) = (x - 3)^2(x^2 - x - 1)$. $x = 3$ is a double root of $P(x) = 0$.
If $(x - a)^3$ is a factor of a polynomial $P(x)$, then $x = a$ is a **triple root** of $P(x) = 0$.	$A(x) = (x + 1)^3(x - 2)$ $x = -1$ is a triple root of $A(x) = 0$.

Example 13

Sketch $P(x) = -(x-3)^3(x+1)$.

Solution

Solve $P(x) = 0$ to find the x-intercepts.

$\therefore x = 3$ and $x = -1$

$x = 3$ is a triple root and $x = -1$ is a single root, so the graph will cross the x-axis at $x = 3$ (and be flat similar to $y = x^3$ there) and at $x = -1$.

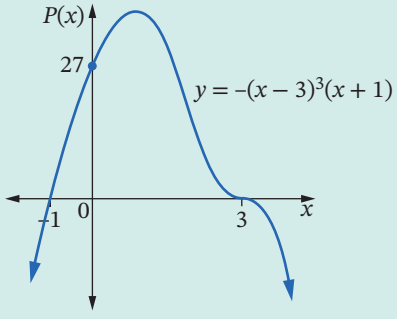
Substitute $x = 0$ to find the y-intercept.

$y = -(0-3)^3(0+1)$

$= 27$

The polynomial is a quartic (degree 4), the leading term is $-x^3 \times x = -x^4$ so the leading term is -1, which is negative, which means the quartic curve will decrease at the end.

Use graphing technology to check the shape of this graph.



Roots of a polynomial equation

- If $P(x) = 0$ has a **single root** at $x = a$, then the graph of the polynomial **crosses** the x-axis at $x = a$.
- If $P(x) = 0$ has a **double root** at $x = a$, then the graph of the polynomial **touches** the x-axis at $x = a$ with a flat gradient, taking the shape of a parabola's vertex there.
- If $P(x) = 0$ has a **triple root** at $x = a$, then the graph of the polynomial **crosses** the x-axis at $x = a$ with a flat gradient, taking the shape of a cubic curve there.

EXERCISE 14.08 ANSWERS ON P. 686

Graphing polynomials U F PS R C

Graphing technology can be used to check the shapes of the graphs in this exercise.

1 Sketch the graph of each polynomial. PS R C

EXAMPLE 12

a $P(x) = x^3 + 6x^2 + 8x$

b $P(x) = x^3 + 2x^2 - x - 2$

c $P(x) = 2x^3 + 3x^2 - 18x + 8$

d $P(x) = 2x^4 - 5x^3 + 2x^2$

e $P(x) = 2x^3 + 17x^2 + 31x - 20$

f $P(x) = -x^3 + 5x^2 + 8x - 12$

g $P(x) = -x^4 + 3x^3 + 6x^2 - 8x$

h $P(x) = -x^4 + 13x^2 - 36$

2 Sketch the graph of each polynomial. PS R C

EXAMPLE 13

a $y = (x-2)(x+4)^2$

b $y = -(x+2)(x-3)^2$

c $y = -(x+1)^2(x-2)^2$

d $y = x(x+1)(x-3)^2$

e $y = -(x-1)(x-2)(x+2)^2$

f $y = -x(x-4)^2$

g $y = -(x-2)^2(x+1)^2$

h $y = (x-4)^3(x+1)$

i $y = (x^2-4)(x-4)^2$

j $y = -x^3 - x^2 + 5x - 3$

k $P(x) = x^4 - 5x^2 + 4$

l $y = -x^4 - 3x^3 - 12x^2 - 44x - 48$

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PATH

3 A monic cubic polynomial has a double zero at $x = -1$ and a single zero at $x = 2$. R C

a Write the equation of this polynomial.

b Hence sketch its graph.

4 **a** What is the maximum number of zeros that a quadratic polynomial $ax^2 + bx + c$ can have?

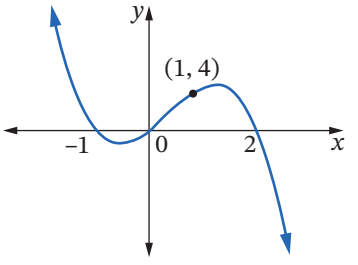
b What is the maximum number of zeros that a cubic polynomial $ax^3 + bx^2 + cx + d$ can have?

c What is the maximum number of zeros that a quartic polynomial $ax^4 + bx^3 + cx^2 + dx + e$ can have?

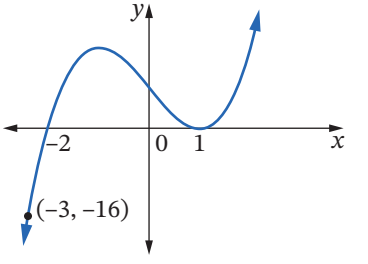
d What is the maximum number of zeros that a polynomial of degree n can have? R C

5 Determine the equation of each cubic curve, in factored form. R C

a



b



6 **a** Explain why $P(x) = (x + 2)(x^2 - 3x + 7)$ has only 1 zero. You may use calculations to support your answer. R C

b How many times does the graph of this polynomial cut the x-axis? Justify your answer.

7 Justify that $G(x) = (x + 1)(x^2 - x + 3)$ has only 1 zero. R C

Justify means to support an argument or conclusion. In Mathematics, it also means to use calculations or graphs to explain your answer.

Technology

Transforming graphs

1 Use graphing technology to graph $P(x) = x(x - 1)(x + 1)$.

2 Write the equations of each polynomial.

a $-P(x)$ **b** $P(-x)$ **c** $P(x) - 3$ **d** $2P(x)$

3 Graph the polynomials shown in question **2**.

4 Describe how the graphs of the polynomials $y = -P(x)$, $y = P(-x)$, $y = P(x) - 3$, $y = 2P(x)$ can be drawn from the graph of $y = P(x)$.

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Transforming graphs of polynomials 14.09

Transformations of polynomials

Compared to the graph of the polynomial $y = P(x)$, the graph of:

- $y = -P(x)$ is a reflection of $y = P(x)$ in the x-axis
- $y = P(-x)$ is a reflection of $y = P(x)$ in the y-axis
- $y = P(x) + c$ is a vertical translation of the graph $y = P(x)$ shifted up or down
- $y = kP(x)$ is $y = P(x)$ either 'stretched' or 'compressed' vertically, with the same x-intercepts

PATH



Worksheet
Advanced graphs

14.09

Example 14

The graph of $y = P(x)$ is shown.

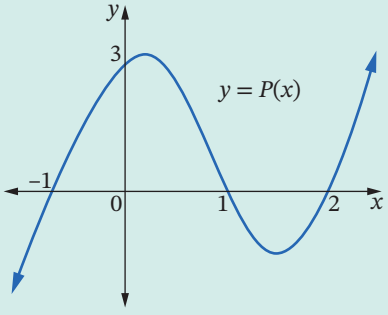
Draw the graphs of each polynomial.

a $y = -P(x)$

b $y = P(-x)$

c $y = P(x) + 2$

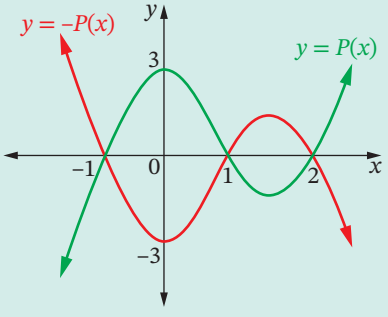
d $y = 3P(x)$



Solution

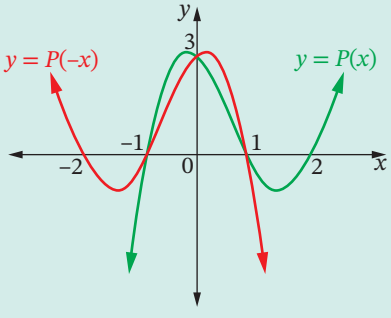
a $y = -P(x)$ is a reflection of $y = P(x)$ in the x-axis.

For each point on $y = P(x)$, the y-value changes sign.



b $y = P(-x)$ is a reflection of $y = P(x)$ in the y-axis.

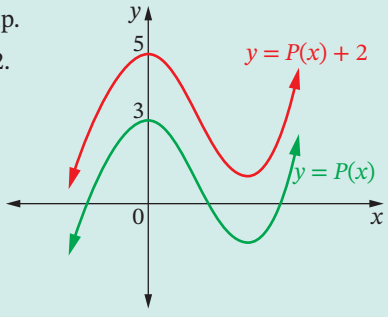
For each point on $y = P(x)$, the x value changes sign.



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c $y = P(x) + 2$ is $y = P(x)$ vertically translated 2 units up.

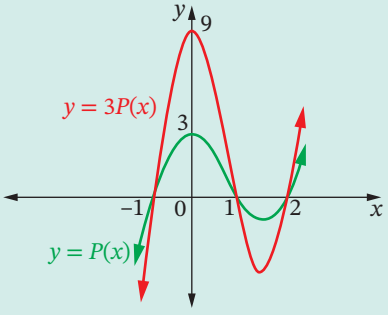
For each point on $y = P(x)$, the y-value increases by 2.



d $y = 3P(x)$ is $y = P(x)$ stretched or dilated vertically by a factor of 3.

For each point on $y = P(x)$, the y-value is multiplied by 3.

A stretch or compression of a graph is called a **dilation**.

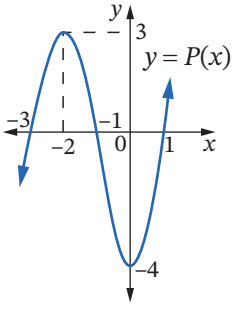


EXERCISE 14.09 ANSWERS ON P. 687

Transforming graphs of polynomials U F R C

EXAMPLE 14

1 The graph of the polynomial $y = P(x)$ is shown.



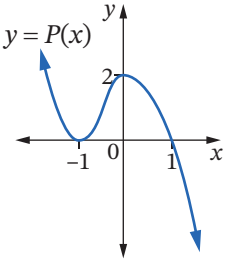
Sketch the graph of each polynomial. R C

a $y = 2P(x)$ **b** $y = P(x) + 2$

c $y = \frac{1}{2}P(x)$ **d** $y = P(x) - 3$

e $y = -P(x)$ **f** $y = P(-x)$

2 Use the graph of $y = P(x)$ shown to sketch each polynomial. R C



a $y = P(x) + 1$ **b** $y = P(x) - 2$

c $y = 2P(x)$ **d** $y = -P(x)$

e $y = P(-x)$ **f** $y = -3P(x)$

3 Graph $P(x) = (x - 1)(x - 3)^2$ and use it to sketch the graph of each polynomial. R C

a $y = -P(x)$ **b** $y = P(-x)$ **c** $y = P(x) - 3$ **d** $y = 2P(x)$

Foundation Standard Complex

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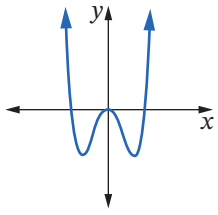
4 Describe the transformation needed to graph each cubic equation using the graph of $y = x^3$. R C

a $y = x^3 - 2$ **b** $y = x^3 + 1$ **c** $y = 2x^3$

d $y = -x^3 + 3$ **e** $y = -3x^3$ **f** $y = 2 - x^3$

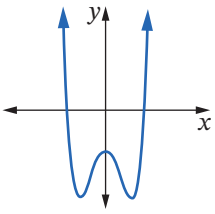
g $y = 2x^3 - 5$ **h** $y = 4 - 3x^3$ **i** $y = \frac{1}{2}x^3 + 4$

5 This is the graph of $y = P(x)$.

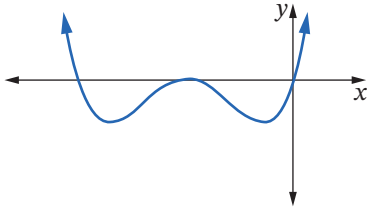


Which of these is the graph of $y = P(x - 2)$? Select **A**, **B**, **C** or **D**.

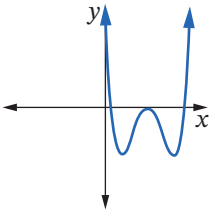
A



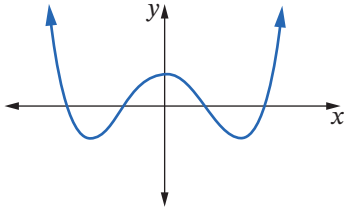
B



C

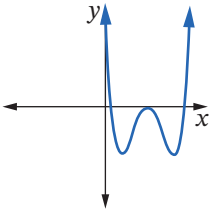


D

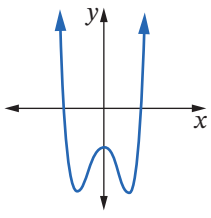


6 Which of these is the graph of $y = P(x) - 2$? Select **A**, **B**, **C** or **D**.

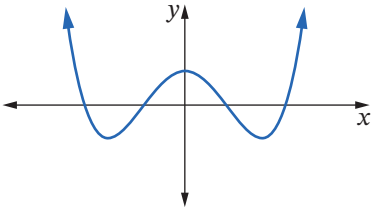
A



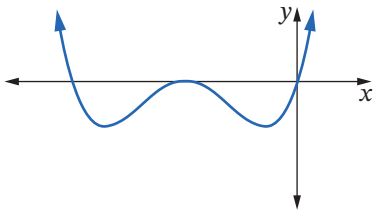
B



C



D



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